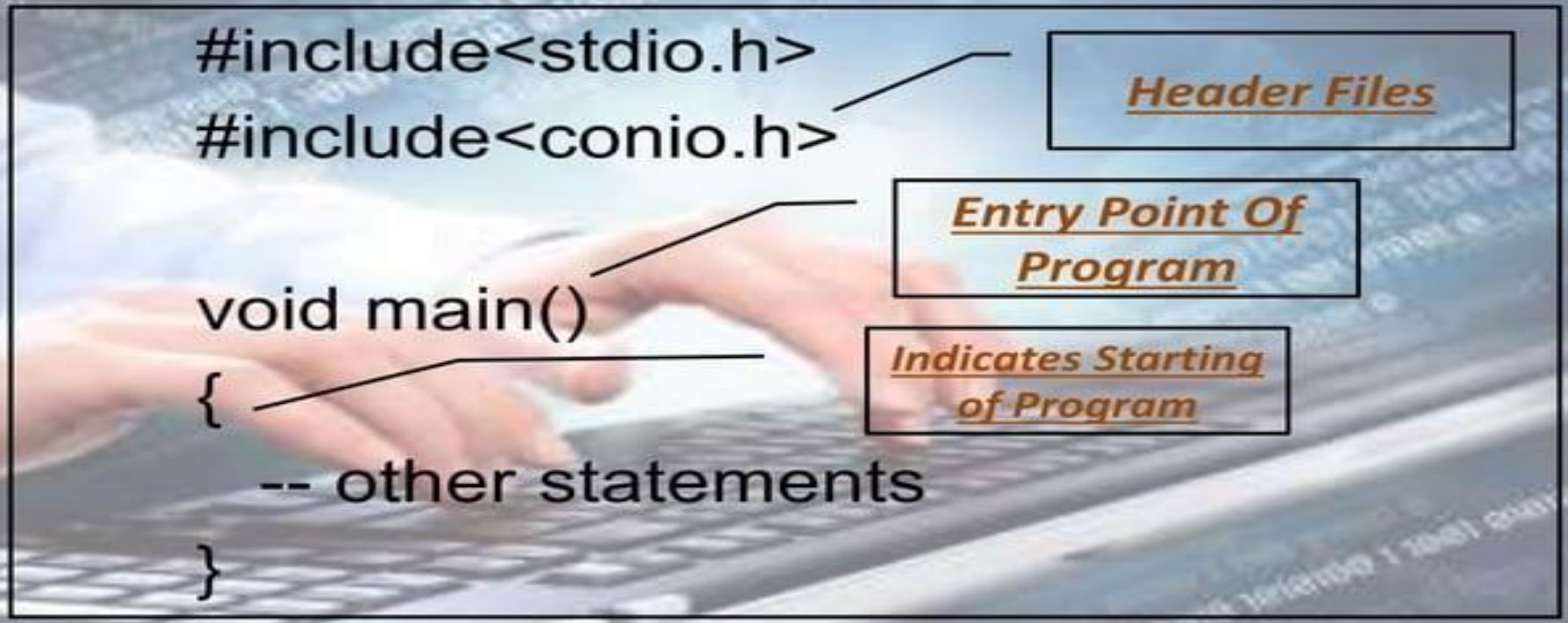
Introduction to C
Basic Input and Output Operations

Basic Structure Of "C" Programs



What is C Language?

C language has standard libraries that allow input and output in a program.

The **stdio.h** or **standard input output library** in C that has methods for input and output.

Output

printf()

To output a string

```
void main()
{
    printf("Hello World!");
}
```

The printf() method, in C, prints the value passed as the parameter to it, on the console screen.

Syntax:

```
printf("%X", variableOfXType);
```

where **%X** is the format specifier in C.

Input

scanf()

The scanf() method, in C, reads the value from the console as per the type specified and store it in the given address.

Syntax:

```
scanf("%X", &variableOfXType);
```

where **%X** is the format specifier. It is a way to tell the compiler what type of data is in a variable and **&** is the address operator in C, which tells the compiler to change the real value of **variableOfXType**, stored at this address in the memory.

How to take input and output of basic types in C?

The basic type in C includes types like int, float, char, etc.
Inorder to input or output the specific type, the **X** in the above syntax is changed with the specific format specifier of that type.
The Syntax for input and output for these are:

Integer:
Input: scanf("%d", &intVariable);
Output: printf("%d", intVariable);

How to take input and output of basic types in C?

Float:
Input: scanf("%f", &floatVariable);
Output: printf("%f", floatVariable);

Character:
Input: scanf("%c", &charVariable);
Output: printf("%c", charVariable);

How to take input and output of basic types in C?

```
#include <stdio.h>
void main()
{
   float num1;
   double num2;
   printf("Enter a number: ");
   scanf("%f", &num1);
   printf("Enter another number: ");
   scanf("%lf", &num2);
   printf("num1 = %f\n", num1);
   printf("num2 = %lf", num2);
}
```

Sample program

```
#include <stdio.h>
void main()
{
/* Declare the variables*/
int num;
char ch;
float f;
/* --- Integer --- Input the integer */
printf("Enter the integer: ");
scanf("%d", &num);
/* Output the integer */
printf("\nEntered integer is: %d", num);
/*--- Float --- Input the float */
printf("\n\nEnter the float: ");
scanf("%f", &f);
/* Output the float */
printf("\nEntered float is: %f", f);
 /* --- Character --- Input the Character*/
printf("\n\nEnter the Character: ");
scanf("%c", &ch);
/*Output the Character */
printf("\nEntered character is: %c", ch);
}
```

ASCII Value

```
#include <stdio.h>
void main()
{
   char chr;
   printf("Enter a character: ");
   scanf("%c", &chr);
   printf("You entered %c.\n",chr);
   printf("ASCII value is %d.", chr);
}
```

Advanced type in C

The advanced type in C includes type like String

Syntax:

Input: scanf("%s", stringVariable);

Output: printf("%s", stringVariable);

Example:

```
void main()
{
   char str[50];
   printf("Enter the Word: ");
   scanf("%s\n", str);
   printf("\nEntered Word is: %s", str);
}
```

To read a full sentence including space

scanf(“%[^\n]s”, string);

Format Specifiers in C

- The format specifier in C -tell the compiler about the type of data to be printed or scanned in input and output operations.
- Start with a **%** symbol and are used in the formatted string in functions like printf(), scanf, sprintf(), etc.

Format specifier	Description
%c	For character type.
%d	For signed integer type.
%f	For float type.
%i	signed integer
%ld or %li	Long

Format specifier	Description
%lf	Double
%Lf	Long double
%s	String
%u	Unsigned int

I/O Multiple Values

```
#include <stdio.h>
void main()
{
  int a;
  float b;
  printf("Enter integer and then a float: ");
  scanf("%d%f", &a, &b);
  printf("You entered %d and %f", a, b);
}
```

I/O Multiple Values

```
#include <stdio.h>

void main() {
   /* using printf() for multiple outputs */
   int day = 07;
   int month = 08;
   int year = 2024;

   printf("The date is: %d-%d-%d", day, month, year);

}
```

Simple calculations inside printf()

```
#include <stdio.h>
void main()
{
    int a = 5, b = 6;
    printf("%d", a + b);

}
```

Multiple Inputs from the User

```
#include <stdio.h>
void main()
{
   /* using scanf() for multiple inputs*/
   char gender;
   int age;

   printf("Enter your age and then gender(M, F or O): ");
   scanf("%d %c", &age, &gender);
   printf("You entered: %d and %c", age, gender);

}
```